

Amazon E-Commerce Analytics Report

Objective:

This project aims to analyze e-commerce sales data to understand revenue trends, customer behavior, product performance, and the impact of discounts and marketing strategies. The project includes data cleaning, feature engineering, visualization, and dashboard creation using Google Sheets and Looker Studio.

Dataset overview:

The dataset used in this project is an e-commerce dataset containing over 45,000 rows and multiple product, sales, and customer-related attributes. The dataset includes important fields such as product name, ratings, review count, monthly sales, pricing information, bestseller status, sponsored status, and coupon availability.

To improve analytical understanding and business insights, several engineered features were created, including Discount Percentage, Discount Category, Estimated Revenue, Potential Revenue, Revenue Loss, and Rating Category.

The dataset was used to analyze various business questions such as:

- How reviews influence product sales
- Whether sponsored products perform better than organic products
- The impact of coupons and discounts on revenue
- Revenue contribution by rating category
- Performance comparison of bestseller and non-bestseller products

These features and analyses helped in identifying customer behavior patterns, product performance trends, and revenue-related insights.

Cleaning Process:

The data cleaning process began with identifying inconsistencies and formatting issues across different columns in the dataset.

The product title column contained excessively long and unstructured product names that did not follow a consistent naming pattern, making standardization difficult during the initial cleaning phase.

The review rating column contained ratings embedded within text sentences rather than numerical values. To extract the rating values, the REGEXEXTRACT function was used to isolate the required numeric pattern.

The review count column was originally stored as text data. This column was converted into numeric format using the VALUE function to enable proper quantitative analysis.

Similarly, the last month sales column also contained sales values embedded within text strings. In addition, some values included the “K” notation representing thousands. To solve this:

- REGEXMATCH was used to detect the presence of “K”
- REGEXEXTRACT was used to isolate the numeric value
- The extracted value was multiplied by 1000 where required
- IFERROR was used to handle potential extraction and conversion errors

Using similar cleaning and transformation techniques, the remaining columns were standardized and converted into analysis-ready formats.

Additional engineered features such as Discount Percentage, Discount Category, Estimated Revenue, Potential Revenue, Revenue Loss, and Rating Category were also created to improve business understanding and analytical depth.

KPI

Several key performance indicators (KPIs) were created to better understand revenue performance, customer behavior, and overall product trends within the dataset.

Estimated Revenue was used to analyze the approximate revenue generated by products based on their discounted selling price and sales volume. Since this metric included discounted pricing, an additional KPI called Potential Revenue was created to estimate the revenue that could have been generated using the original product price before discounts.

To further analyze the impact of discounts, Revenue Loss was calculated to measure the difference between Potential Revenue and Estimated Revenue. This helped in understanding how much potential revenue was reduced due to discount strategies.

Average Rating was used to evaluate overall customer satisfaction and understand the general rating distribution within the dataset. This also helped identify whether the dataset was positively or negatively skewed in terms of customer feedback.

Finally, Total Sales was used to analyze overall product sales performance and estimate customer purchasing activity across the platform.

Analytics & Insights

KPI Overview:

The analysis identified approximately 1.6B in estimated revenue and nearly 1.8B in potential revenue based on original product pricing. The difference between these values resulted in an estimated revenue loss of approximately 172.7M due to discount strategies applied across products.

In addition, the dataset maintained a high average customer rating of 4.4 and total estimated sales of approximately 42.7K, indicating strong customer engagement and product demand.

Revenue Analysis:

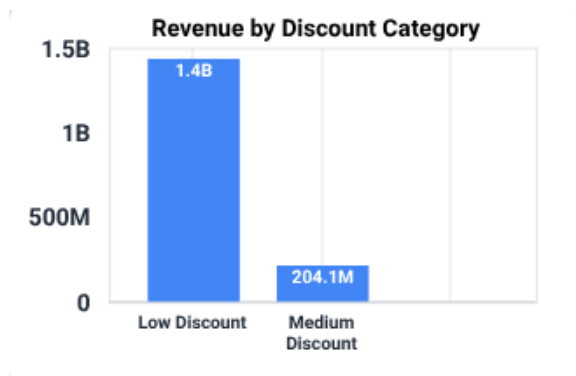


Figure 2: Revenue by Discount Category



Figure 1: Revenue Loss by Discount Category

The revenue performance analysis showed that products within the low discount category generated approximately **1.4B** in estimated revenue, while the medium discount category generated approximately **204.1M**.

This suggests that products with lower discounts were still able to achieve significantly higher sales performance. One possible reason could be that customers were purchasing highly demanded or essential products such as laptops, batteries, and mobile phones, where purchasing decisions may depend more on product value and necessity rather than discount percentage alone.

Further analysis of revenue loss showed that the low discount category contributed approximately **70.6M** in revenue loss, while the medium discount category contributed nearly **102.1M** in revenue loss.

Interestingly, despite generating lower overall revenue, the medium discount category resulted in higher revenue loss. This may indicate that increasing discount percentages did not proportionally improve revenue generation and may have reduced overall revenue efficiency.

Customer Behaviour Analysis:

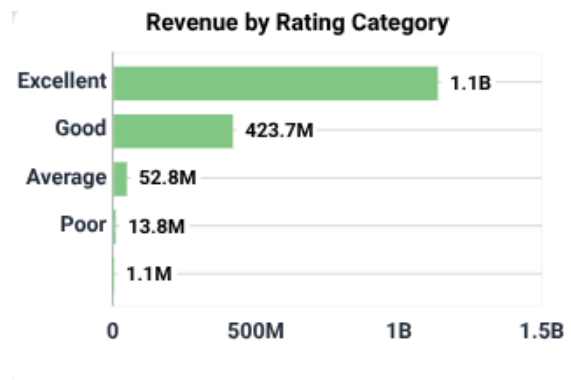


Figure 3: Revenue by Rating Category

The dataset maintained an average customer rating of 4.4, indicating a highly positively skewed rating distribution. This suggests that most customers were generally satisfied with product quality, delivery experience, and overall platform services.

Further analysis of revenue by rating category revealed a strong relationship between customer ratings and revenue performance. Products categorized under the Excellent rating category generated approximately 1.1B in revenue, significantly outperforming other categories.

The Good rating category generated approximately 423.7M in revenue, followed by the Average category with 52.8M and the Poor category with approximately 13.8M.

A clear pattern can be observed where revenue consistently decreases as rating categories decline. This may indicate that lower-rated products are potentially affected by issues such as product quality concerns, delayed deliveries, exchange or return problems, or negative customer service experiences.

The analysis suggests that customer satisfaction plays an important role in revenue generation and long-term product performance.

Marketing & Promotional Analysis

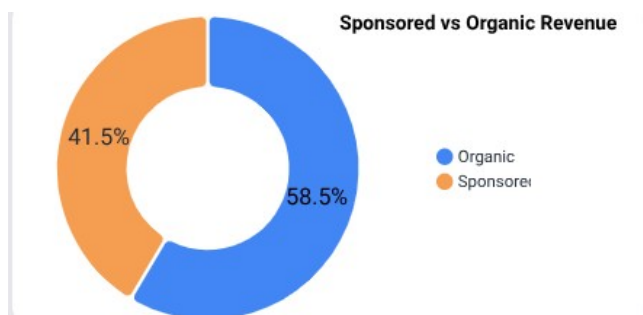


Figure 5: Sponsored vs Organic Revenue

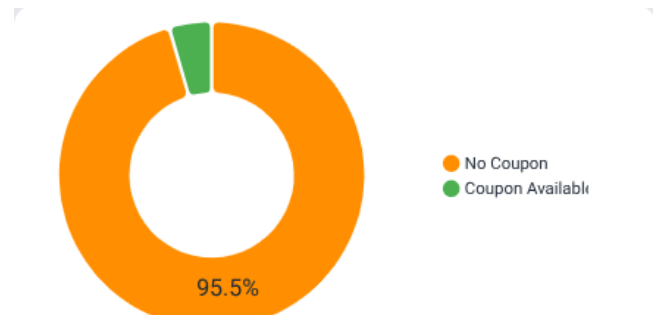


Figure 4: No Coupon vs Coupon Available

The marketing and promotional analysis shows that organic products contributed approximately 58.5% of total revenue, while sponsored products generated around 41.5%. This indicates that organic listings performed better overall in terms of revenue generation. One possible reason is that customers may trust organically performing products more, especially for high-value categories such as laptops and smartphones, where buyers often rely on product reputation, ratings, and reviews rather than sponsored visibility alone.

In the coupon analysis, nearly 95.5% of revenue was generated from products without coupons, while only 4.5% came from coupon-based purchases. This suggests that coupon usage in the dataset was relatively low and that most customers purchased products without relying heavily on discount coupons.

Product Performance Analysis

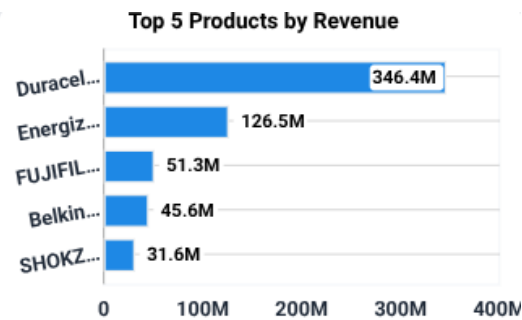


Figure 6: Top 5 Products by Revenue

Product performance analysis reveals that the Duracell CopperTop 9V Battery was the highest revenue-generating product, contributing approximately 346.4M in revenue. The exceptionally high revenue contribution may indicate strong recurring demand, high purchase frequency, and broad consumer utility, as battery products are commonly used across multiple electronic devices and household applications.

The second highest-performing product was Energizer Alkaline Power AAA Batteries, generating approximately 126.5M in revenue. The dominance of battery-related products among top-performing items suggests that essential utility products with frequent replacement cycles contribute significantly to overall sales performance.

Fujifilm products generated approximately 51.3M in revenue, while Belkin MagSafe accessories contributed around 45.6M. This indicates strong consumer demand for electronic accessories and mobile device-related products, likely driven by increasing dependence on smartphones, portable devices, and digital ecosystems.

Additionally, Shokz OpenRun Pro 2 generated approximately 31.6M in revenue, showing notable customer interest in premium audio and wearable technology products. The presence of tech accessories and premium electronics among top-performing products may suggest that consumers are willing to spend more on convenience-driven and lifestyle-enhancing technology products.

Dataset Limitations

1. Missing Shipping Date Information

A significant number of null values were observed in the shipping date column, which limits the ability to perform accurate shipping performance and delivery time analysis. Due to missing shipping records, customer delivery behavior and logistics efficiency could not be fully evaluated.

2. Positively Skewed Revenue Distribution

The dataset appears to be positively skewed, where a small number of products contributed disproportionately high revenue compared to the majority of products. This may affect average-based interpretations, as high-performing products can heavily influence overall revenue metrics.

3. Limited Coupon Usage Data

The dataset contains relatively low coupon-based transactions compared to non-coupon purchases. As a result, coupon effectiveness and promotional impact analysis may not fully represent customer discount behavior.

4. Lack of Customer Demographic Information

The absence of customer demographic data such as age, gender, or location limits deeper customer segmentation and behavioral analysis.

Recommendation

Since organic products generated higher revenue compared to sponsored products, the company should focus more on improving sponsored advertisement strategies and product targeting. This may help improve conversion rates and customer engagement for promoted products.

Battery and electronic utility products contributed a major portion of total revenue, indicating strong customer demand in essential and frequently used product categories. The company can further expand these categories and maintain proper stock availability to avoid supply shortages.

Coupon-based purchases contributed very little revenue compared to non-coupon purchases. This may indicate limited coupon availability or lower customer usage of discount offers. The company can test targeted promotional campaigns during seasonal sales or festivals to improve customer participation.

The dataset also contained missing shipping date values, which limited detailed delivery and logistics analysis. Improving shipment tracking and maintaining complete operational records may help the company perform better delivery performance analysis in the future.

Additionally, collecting more customer-related information such as location, demographics, and buying behavior may help the company perform deeper customer segmentation and personalized marketing analysis.

Conclusion

The analysis shows that electronic accessories, utility products, and battery-related products contributed significantly to overall revenue generation. Organic products generated higher revenue compared to sponsored products, indicating stronger customer preference toward naturally performing products rather than promoted listings. The dataset also highlights that highly rated products contributed a major portion of revenue, suggesting that customer satisfaction and product quality play an important role in sales performance. Additionally, coupon-based purchases contributed relatively low revenue, which may indicate limited promotional usage within the dataset. Through data cleaning, visualization, and dashboard analysis, several important business insights related to product performance, customer behavior, and promotional effectiveness were identified. However, some limitations such as missing shipping date values and incomplete operational data restricted deeper logistics and customer behavior analysis.

Overall, the analysis provides useful insights that can help improve product strategy, promotional planning, and business decision-making.